DATABASE MANAGEMENT SYSTEM

Data vs. information:

DATA

- Data is a *collection of facts*, such as values or measurements.
- It can be numbers, words, measurements, observations or even just descriptions of things.

INFORMATION

- Information is data that have been organized and communicated in a *coherent and meaningful manner*.
- Data is converted into information, and information is converted into knowledge.
- Knowledge; information evaluated and organized so that it can be used purposefully.

Data is also stored in excel sheets.

Is it different from database?

DATABASE

A collection of related pieces of data:

- Representing/capturing the information about a real-world enterprise or part of an enterprise.
- A database is designed ,built ,populated with data for a *specific purpose.*

  It has intended group of users and preconceived application.
- Activities of the enterprise are supported by the database and continually update the database.

Example: University Database

- Data about students, faculty, courses, laboratories, course registration/enrollment etc.
- Purpose: To keep an accurate track of the academic activities of the university.

What is a DBMS?

A ***Database Management System (DBMS)*** *is a* software package designed to store and manage databases.

DBMS tasks:

- Managing large quantity of **structured data**
- **Efficient retrieval and modification:** query processing and optimization
- **Sharing data:** multiple users use and manipulate data
- **Controlling the access to data**: maintaining the data integrity

Database Applications:

- Banking: all transactions
- Airlines: reservations, schedules
- Universities: registration, grades
- Sales: customers, products, purchases
- Manufacturing: production, inventory, orders, supply chain
- Human resources: employee records, salaries, tax deductions

Key Functions of a DBMS

1. **Data Storage:**

- DBMS stores data in organized structures, typically in the form of tables with rows and columns.
- Ensures data is stored efficiently and securely.

2. **Data Retrieval:**

- DBMS allows users to query the database using SQL (Structured Query Language) to fetch specific data quickly.
- Uses indexes and optimized storage techniques to speed up retrieval, even with large amounts of data.

3. Data Manipulation:

- Users can insert, update, and delete data using commands like INSERT, UPDATE, DELETE in SQL.
- Allows batch processing and transactions, ensuring data integrity and consistency during operations.

4. Data Security and Integrity:

- Enforces access controls so only authorized users can view or manipulate data.
- Ensures integrity by enforcing constraints like primary keys, foreign keys, and unique constraints.

5. Backup and Recovery:

Provides automatic backup and recovery features to protect data from accidental loss or corruption.

FILE SYSTEM

What is File System?

A **File System** is a method and structure used by an **operating system** to organize, store, retrieve, and manage files on a storage device such as a **hard drive, SSD, or USB**.

It provides the foundation for storing data in files and directories, but **lacks the features of a Database Management System (DBMS).**

Drawbacks of old File methods

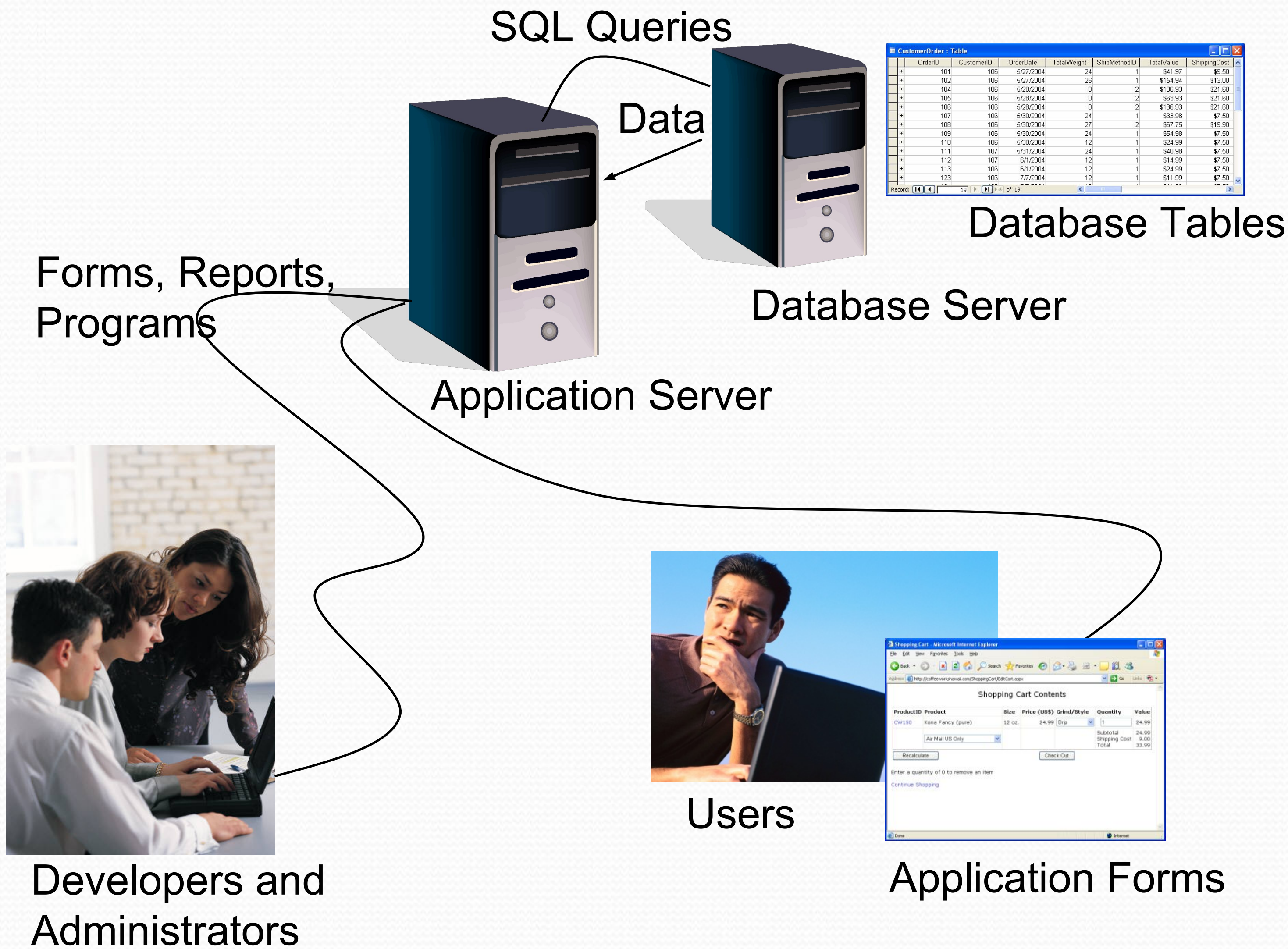
Feature	File System	DBMS
Definition	A system that manages files on a disk or storage device.	A software that manages databases, allowing storage, retrieval, and manipulation of data.
Data Storage	Stores data in files within directories.	Stores data in structured tables using rows and columns.
Data Organization	Unstructured; each file is independent.	Highly structured; data is related through tables, schemas, and relationships.
Redundancy	High redundancy due to lack of structure and relationships.	Minimizes redundancy by using relational models and enforcing constraints.
Data Integrity	No inherent mechanisms to ensure data accuracy or integrity.	Enforces data integrity through constraints like primary keys, foreign keys, and unique constraints.
Security	Basic file-level permissions (read, write, execute).	Fine-grained security controls; can restrict access based on user roles.
Concurrency Control	Minimal support for multiple users accessing files simultaneously.	Advanced concurrency control with mechanisms like locking, time-stamping, and transactions.

Feature	File System	DBMS
Data Querying	No querying capabilities; users access data by opening files.	Supports complex querying with SQL (Structured Query Language).
Data Relationships	No direct support for relationships between data files.	Supports relationships between tables via primary and foreign keys.
Data Independence	Low data independence; changes to file structure may affect applications.	High data independence; changes to database structure do not affect application code.
Backup and Recovery	Manual backup, with limited recovery options.	Automated backup and recovery mechanisms, including point-in-time recovery.
Transactions	Does not support atomic transactions.	Supports atomic transactions (**ACID properties: Atomicity, Consistency, Isolation, Durability**).
Performance	Slower when dealing with large amounts of data due to lack of indexing and optimization.	Optimized for performance with indexing, query optimization, and efficient storage.
Example	File systems like FAT, NTFS, ext4 (e.g., in Windows or Linux).	DBMS like MySQL, Oracle, PostgreSQL, Microsoft SQL Server.

Conclusion:

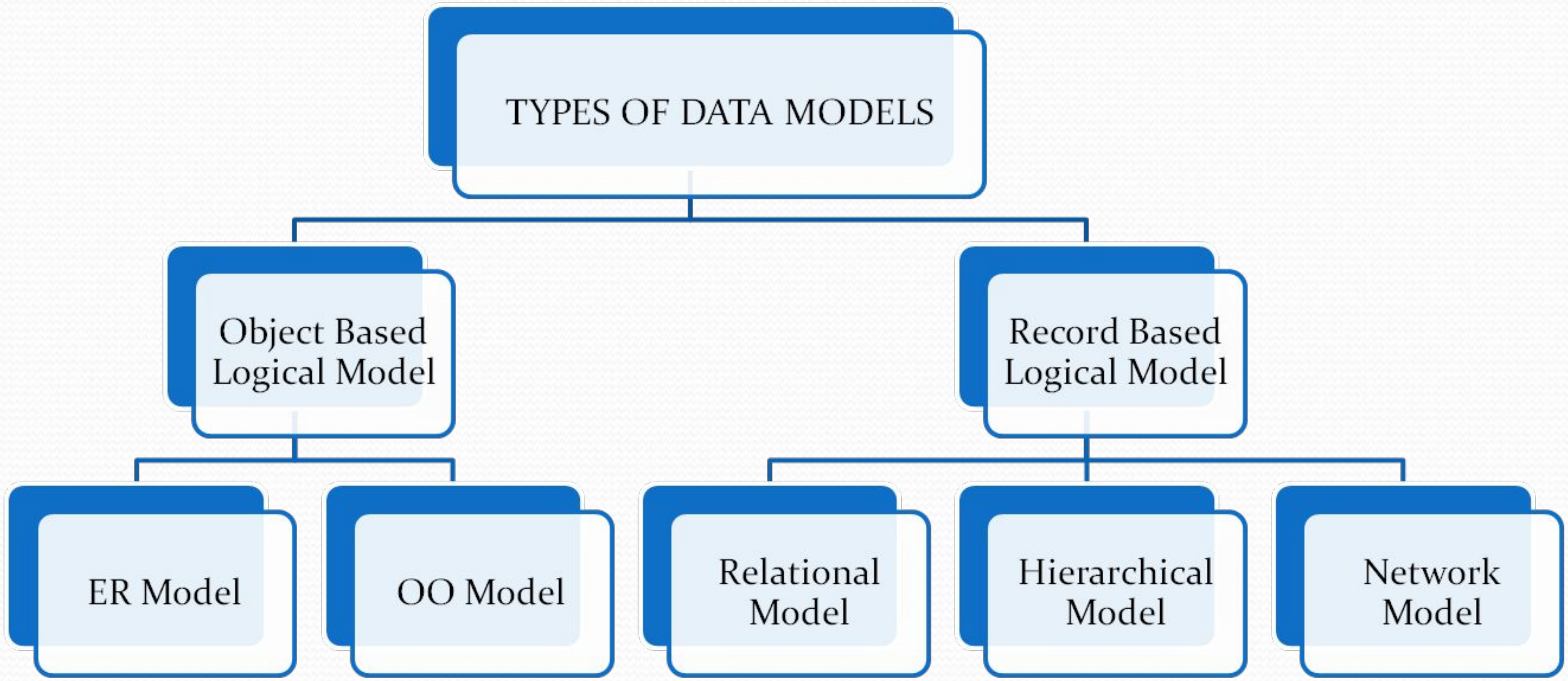
A DBMS is a more advanced system than a File System, especially when handling large and complex datasets that require consistency, security, and efficiency.

Application Development with a DBMS



Data Models

- A **data model** is a collection of concepts(tools and languages) for describing data.
- Data Models define underlying structure of DBMS.
- Contains Description of data, data relationship ,data semantics , data integrity constraints.



Data Models cont.

- The **relational model** of data is the most widely used model today.
  - Main concept: relation, basically a table with rows and columns.
  - Every relation has a schema, which describes the columns, or fields.

Instances and Schemas

- **Schema** – A **schema** is a description of a particular collection of data, using the a given data model.

Schema is the overall design of database.

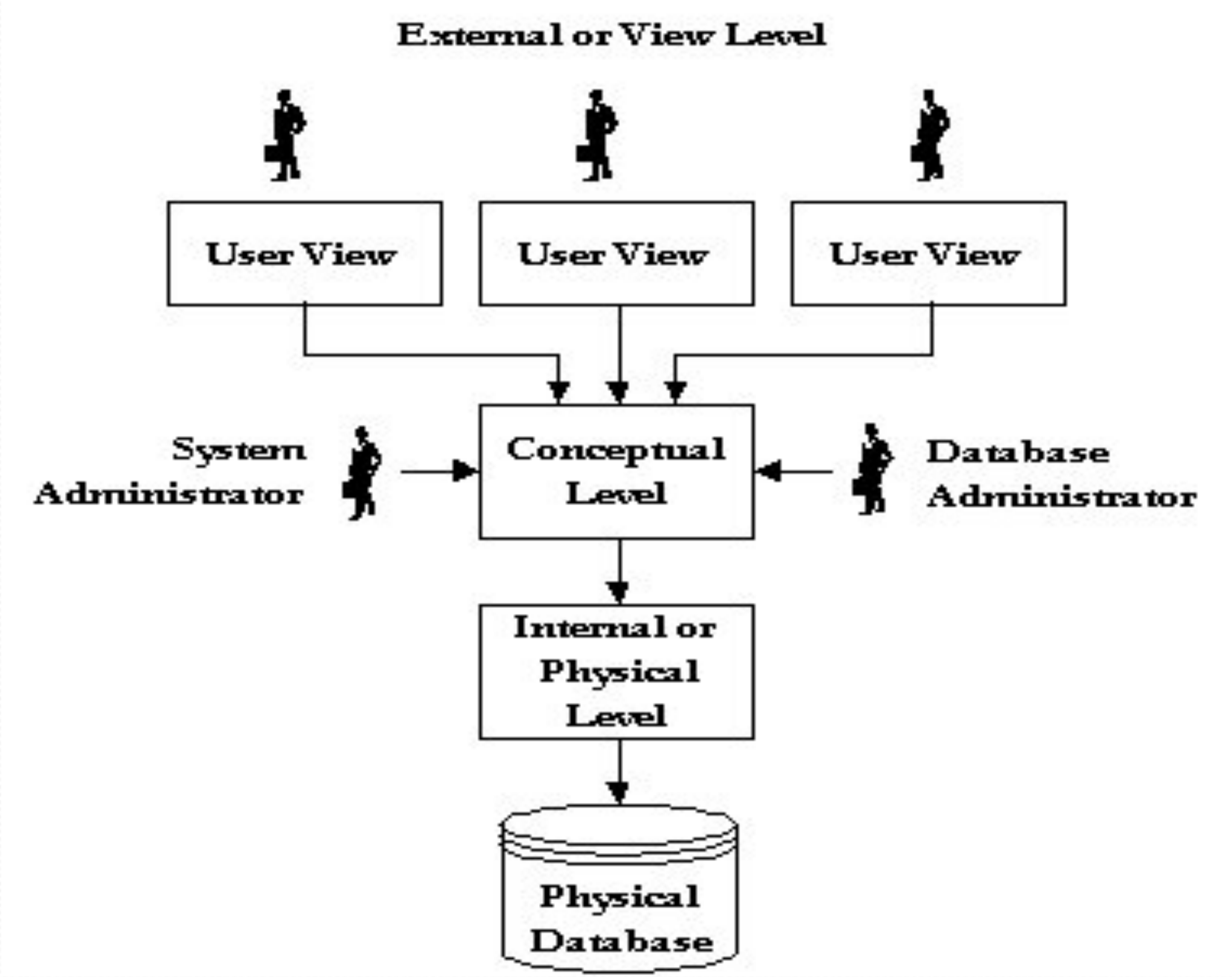
In RDBMS context:

Schema – table names, attribute names with their data types for each table and constraints etc.

Name	Roll	Class	Subject

- Data in a database at particular moment is called *database state or snapshot.*
- It is sometimes called the current state or ***instances*** in database***. Eg.*** Entry in student table.
- DBMS stores description of schema construct and constraints known as ***Metadata.***
- The Schema is sometimes called ***Intension*** and instance is called ***extension*** of schema

Levels of Abstraction: Three-schema Architecture



- **Physical level:**
  - describes details of how data is stored: files, indices, etc. on the random access disk system
  - It also typically describes the record layout of files and type of files (hash, b-tree, flat).

- **Conceptual level(logical):**
  - Describes data stored in database, and the relationships among the data.
  - Hides details of the physical level.
  - In the relational model, the conceptual schema presents data as a set of tables

    type customer = **record**
    name : string;
    street : string;
    city : integer;
    end;

- **View level :**
  - Each view describes an aspect of the database relevant to a particular group of users.
  - Portions of stored data should not be seen by some users and implement a level of security .

    For instance, in the context of a library database:
    - Books Purchase Section
    - Issue/Returns Management Section

Three Level Architecture

External Level

Doctor's View

SSN | *dosage*

Billing Office View

SSN | *LastName* | *FirstName* | ...

Conceptual Level

SSN | *Dosage* | *LastName* | *FirstName* | ...

Internal Level

```
struct {
   string SSN;
   double dosage;
   string LastName;
   string FirstName;
   ...
   }
```

Roles for people

- **Application programmers** – interact with system through DML calls
- **Sophisticated users** – form requests in a database query language
- **Specialized users** – write specialized database applications that do not fit into the traditional data processing framework
- **Naive users** – invoke one of the permanent application programs that have been written previously
  - E.g. people accessing database over the web, bank tellers, clerical staff

DBA (Database Administrator)

- Designing the logical schema
- Creating the structure of the entire database
- Monitor usage and create necessary index structures to speedup query execution
- Grant / Revoke data access permissions to other users etc

Data Independence

Capacity to change schema at one level of database system without having to change schema at next higher level.

- ***Physical data independence:*** *The ability to modify physical level schema without affecting the logical or view level schema.*

Performance tuning – modification at physical level creating a new index etc.

Logical data independence: *The ability to change the logical level scheme without affecting the view level schemes or application programs*

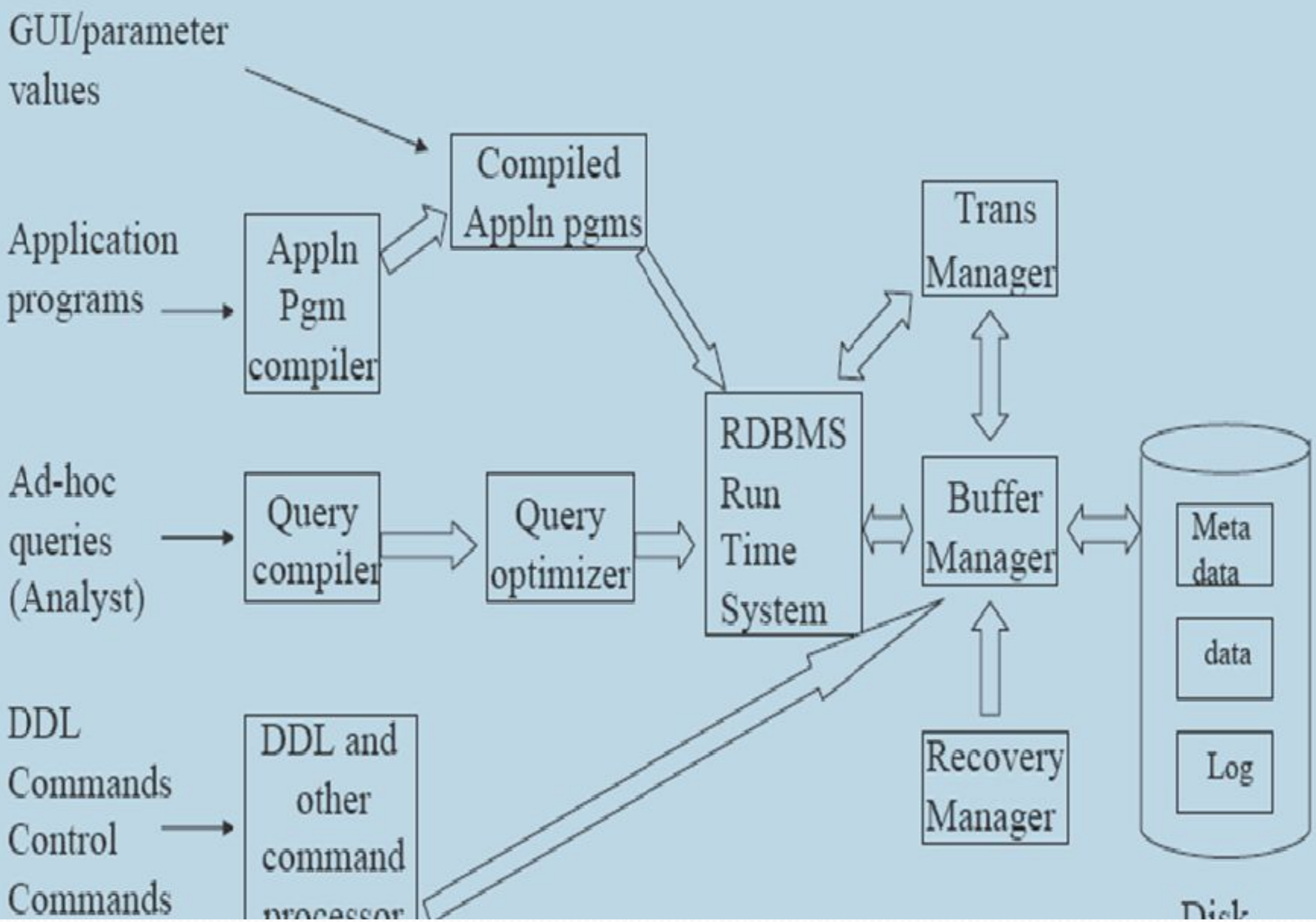
Adding a new attribute to some relation

- no need to change the programs or views that don’t require to use the new attribute

Deleting an attribute

- no need to change the programs or views that use the remaining data

Architecture of an RDBMS system



Summary

- DBMS used to maintain, query large datasets.
- Benefits include recovery from system crashes, concurrent access, quick application development, data integrity and security.
- Levels of abstraction give data independence.
- A DBMS typically has a layered architecture.